

## Evaluation frame



**294 pages**

229 government + 65 arXiv



**Held-out Q-A probe**

**663 Q-A**

527 government + 136 arXiv

query + reference page  
+ answer span

## Candidate pipelines

Parser candidates  $P_i$

$P_1$   $P_2$  ...  $P_m$

Parsed page outputs

Intrinsic  
score

Chunker candidates  $C_j$

$C_1$   $C_2$  ...  $C_c$

Intrinsic  
ranking

all  $P_i \times C_j$  pairs

Compare  
Rankings  
(same chunker)

C1

Intrinsic  
ranking  
can mislead

Candidate corpora

$C_j(P_i)$ ,  $m \times c$

Index each corpus

candidate index

## RCPS evaluation

Search with probe queries

BGE-M3

mE5-large

Qwen3-Emb-8B

Relevant chunk

from reference page AND contains  
normalised answer span

MRR@k

RCPS(P,C)

mean MRR across retrievers and cutoffs

C3

Controlled settings

Probe set

Retriever models

Cutoffs  $k = 1, 5, 10$

Relevance rule

Candidate-wise RCPS scores

one score for every  $P_i \times C_j$

## Select, diagnose + act

RCPS score matrix

|       | $C_1$ | ... | $C_j$ | ... | $C_c$ |
|-------|-------|-----|-------|-----|-------|
| $P_1$ |       |     |       |     |       |
| $P_i$ |       |     | ✓     |     |       |
| $P_m$ |       |     |       |     |       |

RCPS rank | latency/compute tie-break

Selected  $P^* + C^*$

parser + chunker configuration

$P^*$  output

$C^*$  chunks

Ref. page  
+ span

Coverage diagnostic

retriever-free exact-span check

C2

Absent

no match  
in output

Covered

one chunk

Split

across chunks

Inspect /  
switch P

Proceed

Overlap /  
rechunk

Parser training  
(optional)

C4

Re-evaluate  
changed P/C

Deploy final  $P^* + C^*$